

Matrix Analysis Final

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Problem 1.

Let $p = (p_1, \dots, p_n)^T \in \mathbb{R}^n$ with $p_i > 0$ and $\sum_{i=1}^n p_i^2 = 1$. Set

$$S(p) = \sum_{i=1}^n \left(p_i + \frac{1}{p_i} \right).$$

We prove

$$n^3 + 2n + \frac{1}{n} \leq S(p)^2.$$

First find the minimum of $S(p)$ under the constraint $\sum_{i=1}^n p_i^2 = 1$. Introduce the Lagrangian

$$\mathcal{L}(p, \lambda) = \sum_{i=1}^n \left(p_i + \frac{1}{p_i} \right) + \lambda \left(\sum_{i=1}^n p_i^2 - 1 \right).$$

Differentiating w.r.t. p_i gives, for all i ,

$$1 - \frac{1}{p_i^2} + 2\lambda p_i = 0.$$

The right-hand side is the same function of p_i for every i , so at a critical point all p_i 's are equal: $p_i = c > 0$. Using $\sum p_i^2 = 1$ gives $nc^2 = 1$, hence $c = 1/\sqrt{n}$.

On the boundary (some $p_i \rightarrow 0$) the sum $S(p) \rightarrow \infty$, so S attains its global minimum at $p_i = 1/\sqrt{n}$ and

$$S_{\min} = n \left(\frac{1}{\sqrt{n}} + \sqrt{n} \right) = \sqrt{n} + n^{3/2}.$$

Therefore

$$S(p)^2 \geq S_{\min}^2 = (\sqrt{n} + n^{3/2})^2 = n^3 + 2n^2 + n.$$

Finally,

$$S(p)^2 - (n^3 + 2n + \frac{1}{n}) = n^3 + 2n^2 + n - (n^3 + 2n + \frac{1}{n}) = 2n^2 - n - \frac{1}{n} = \frac{2n^3 - n^2 - 1}{n} \geq 0$$

for all integers $n \geq 1$. Thus

$$n^3 + 2n + \frac{1}{n} \leq \left(\sum_{i=1}^n \left(p_i + \frac{1}{p_i} \right) \right)^2.$$

Problem 2.

Let $a \in \mathbb{R}^n$. Let $D = \text{diag}((-1)^1, \dots, (-1)^n)$ and $e = (1, \dots, 1)^T$. Write

$$S_o = \sum_{\substack{i=1 \\ i \text{ odd}}}^n a_i, \quad S_e = \sum_{\substack{i=1 \\ i \text{ even}}}^n a_i,$$

and denote $n_o = \lceil n/2 \rceil$ (number of odd indices), $n_e = \lfloor n/2 \rfloor$ (number of even indices).

Then

$$e^T a = S_o + S_e, \quad e^T D a = \sum_{i=1}^n (-1)^i a_i = S_o - S_e,$$

hence

$$|e^T a|^2 + |e^T D a|^2 = |S_o + S_e|^2 + |S_o - S_e|^2 = 2(|S_o|^2 + |S_e|^2).$$

By Cauchy-Schwarz on the odd and even subsets,

$$|S_o|^2 \leq n_o \sum_{i \text{ odd}} |a_i|^2, \quad |S_e|^2 \leq n_e \sum_{i \text{ even}} |a_i|^2.$$

Therefore

$$|e^T a|^2 + |e^T D a|^2 \leq 2 \left(n_o \sum_{i \text{ odd}} |a_i|^2 + n_e \sum_{i \text{ even}} |a_i|^2 \right) \leq 2n_o \sum_{i=1}^n |a_i|^2 = 2 \left\lceil \frac{n}{2} \right\rceil \|a\|_2^2.$$

Problem 3.

Let Z_n be the $n \times n$ shift-up matrix: $(Z_n)_{i,j} = 1$ if $j = i + 1$ and 0 otherwise.

(a) For an $n \times n$ Toeplitz matrix T we consider $C = TZ_n - Z_n T$. For $1 < i < n$, $1 < j < n$,

$$(TZ_n)_{i,j} = T_{i,j-1}, \quad (Z_n T)_{i,j} = T_{i+1,j},$$

and since $T_{i,j} = t_{i-j}$ depends only on $i - j$, these two entries coincide. Thus all interior entries of C are zero; only the first row and last column can be non-zero. Hence C is a sum of at most two rank-one matrices and $\text{rank}(TZ_n - Z_nT) \leq 2$. This bound is attained for a generic Toeplitz matrix, so the maximum rank is 2.

(b) Let V be a unit upper-triangular Toeplitz matrix. Because both V and Z_n have constant diagonals, one checks directly that $VZ_n = Z_nV$. Hence $Z_n = VZ_nV^{-1}$.

(c) Solve $X_RZ_n - Z_nX_R = 0$. Writing $X_R = (x_{ij})$,

$$(X_RZ_n)_{i,j} = x_{i,j-1}, \quad (Z_nX_R)_{i,j} = x_{i+1,j},$$

(with $x_{i,0} = x_{n+1,j} = 0$), so $x_{i,j-1} = x_{i+1,j}$. Thus x_{ij} is constant along each super-diagonal and vanishes below the main diagonal. Hence X_R is an upper-triangular Toeplitz matrix. It is determined by its first row, so the solution space has dimension n .

(d) Similarly, solving $Y_LZ_n^T - Z_n^TY_L = 0$ gives lower-triangular Toeplitz matrices; the solution space again has dimension n .

(e) Consider the linear operator

$$\mathcal{L}(X) = XZ_n - Z_nX.$$

Equip the space of matrices with the Frobenius inner product $\langle A, B \rangle = \text{tr}(A^TB)$. Then

$$\begin{aligned} \langle \mathcal{L}(X), Y \rangle &= \text{tr}((XZ_n - Z_nX)^TY) \\ &= \text{tr}(Z_n^TX^TY) - \text{tr}(X^TZ_n^TY) \\ &= \text{tr}(X^T(Z_n^TY - YZ_n^T)) = \langle X, Z_n^TY - YZ_n^T \rangle. \end{aligned}$$

Hence $\mathcal{L}^*(Y) = Z_n^TY - YZ_n^T$, and

$$\ker \mathcal{L}^* = \{Y : Z_n^TY - YZ_n^T = 0\}$$

is exactly the space of lower-triangular Toeplitz matrices (found in part (d)). By the Fredholm alternative, Y lies in the range of $\mathcal{L}$ iff it is orthogonal to all matrices in $\ker \mathcal{L}^*$, i.e.

$$XZ_n - Z_nX = Y \text{ has a solution } \iff \text{tr}(Y_L^TY) = 0 \text{ for all lower-triangular Toeplitz } Y_L.$$

There are n linearly independent such constraints, matching the null-space dimension of part (c).

(f) To compute a particular solution X_p when Y satisfies the consistency conditions, write

$$(XZ_n - Z_nX)_{i,j} = x_{i,j-1} - x_{i+1,j} = y_{ij},$$

with $x_{i,0} = x_{n+1,j} = 0$. Set $x_{n,j} = 0$ for all j , and then determine the remaining entries recursively upward via

$$x_{i,j-1} = y_{ij} + x_{i+1,j}.$$

This produces a particular solution X_p with $X_pZ_n - Z_nX_p = Y$.

(g) The general solution of $XZ_n - Z_nX = Y$ is

$$X = X_p + X_R,$$

where X_p is any particular solution from (f) and X_R is an arbitrary homogeneous solution, i.e. an upper-triangular Toeplitz matrix from part (c).

(h) Now consider

$$X - Z_n^T X Z_n = Y.$$

Entrywise,

$$(Z_n^T X Z_n)_{ij} = \begin{cases} x_{i-1,j-1}, & i > 1, j > 1, \\ 0, & \text{otherwise,} \end{cases}$$

so the equation is

$$x_{ij} - x_{i-1,j-1} = y_{ij}, \quad 1 \leq i, j \leq n,$$

with $x_{0j} = x_{i0} = 0$. This recurrence uniquely determines x_{ij} if we proceed along anti-diagonals $i + j = k$:

- $x_{11} = y_{11}$,
- $x_{i1} = y_{i1}$ for $i \geq 2$ and $x_{1j} = y_{1j}$ for $j \geq 2$,
- for $i, j \geq 2$, $x_{ij} = y_{ij} + x_{i-1,j-1}$.

Thus the equation always has a unique solution X , and the recurrence above gives an $O(n^2)$ algorithm to compute it.

Problem 4.

Let $-1 < x < 1$ and $n \in \mathbb{N}$. Set

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 2x \end{pmatrix}.$$

Let U_k be the Chebyshev polynomials of the second kind, so that for $x = \cos \theta$ ($0 < \theta < \pi$),

$$U_k(x) = \frac{\sin((k+1)\theta)}{\sin \theta}, \quad k \geq 0,$$

and $U_{k+1}(x) = 2xU_k(x) - U_{k-1}(x)$.

One checks for $n = 1$ that

$$A = \begin{pmatrix} -U_{-1}(x) & U_0(x) \\ -U_0(x) & U_1(x) \end{pmatrix},$$

if we define $U_{-1}(x) := 0$. Using the recurrence and the fact that A satisfies $\lambda^2 - 2x\lambda + 1 = 0$, an induction shows that for all $n \geq 1$,

$$A^n = \begin{pmatrix} -U_{n-2}(x) & U_{n-1}(x) \\ -U_{n-1}(x) & U_n(x) \end{pmatrix}, \tag{1}$$

again with $U_{-1}(x) = 0$.

Let $x = \cos \theta$. Then

$$U_k(x) = \frac{\sin((k+1)\theta)}{\sin \theta}. \tag{2}$$

Take $e_2 = (0, 1)^T$. From (1),

$$A^n e_2 = \begin{pmatrix} U_{n-1}(x) \\ U_n(x) \end{pmatrix},$$

so by (2),

$$\|A^n e_2\|_2^2 = \frac{\sin^2(n\theta) + \sin^2((n+1)\theta)}{\sin^2 \theta} \geq \frac{\sin^2(n\theta)}{\sin^2 \theta}.$$

Hence

$$\|A^n\|_2 \geq \|A^n e_2\|_2 \geq \frac{|\sin(n\theta)|}{|\sin \theta|} = \frac{|\sin(n \arccos x)|}{\sqrt{1-x^2}}.$$

Using $\|\cdot\|_2 \leq \|\cdot\|_F$ and (1),

$$\|A^n\|_2^2 \leq \|A^n\|_F^2 = U_{n-2}(x)^2 + 2U_{n-1}(x)^2 + U_n(x)^2.$$

By (2),

$$\|A^n\|_F^2 = \frac{\sin^2((n-1)\theta) + 2\sin^2(n\theta) + \sin^2((n+1)\theta)}{\sin^2 \theta} \leq \frac{4}{\sin^2 \theta},$$

because each $\sin^2(k\theta) \leq 1$. Thus

$$\|A^n\|_2^2 \leq \frac{4}{\sin^2 \theta} \Rightarrow \|A^n\|_2 \leq \frac{2}{|\sin \theta|} = \frac{2}{\sqrt{1-x^2}}.$$

$$\frac{|\sin(n \arccos x)|}{\sqrt{1-x^2}} \leq \|A^n\|_2 \leq \frac{2}{\sqrt{1-x^2}}.$$

Let $v = (1, x)^T$ and again write $x = \cos \theta$. Define

$$v_k = \begin{pmatrix} \cos((k-1)\theta) \\ \cos(k\theta) \end{pmatrix}.$$

Using $\cos((k+1)\theta) = 2\cos \theta \cos(k\theta) - \cos((k-1)\theta)$, one checks that $Av_k = v_{k+1}$. Since $v_1 = (1, x)^T$, we obtain

$$A^n v = v_{n+1} = \begin{pmatrix} \cos(n\theta) \\ \cos((n+1)\theta) \end{pmatrix}.$$

Hence

$$\|A^n v\|_2^2 = \cos^2(n\theta) + \cos^2((n+1)\theta) \leq 1 + 1 = 2,$$

so

$$\left\| A^n \begin{pmatrix} 1 \\ x \end{pmatrix} \right\|_2 \leq \sqrt{2}.$$